INTERACTIVE TREE SIMULATOR

Interactive simulators for binary search trees, AVL trees & traversal practice

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Why Simulate Trees?

Tree structures — binary trees, BSTs, and AVL trees — are hard to grasp from static diagrams alone. Interactive simulators let learners insert, delete, search, and traverse nodes step by step, watching rotations and rebalancing happen in real time.

1 **Build**

Insert & delete nodes interactively

2 **Visualize**

Watch rotations, balancing, and pointers update live

3 **Traverse**

Practice pre/in/post-order & BFS click-by-click

4 **Test**

Search and sort operations with animated feedback

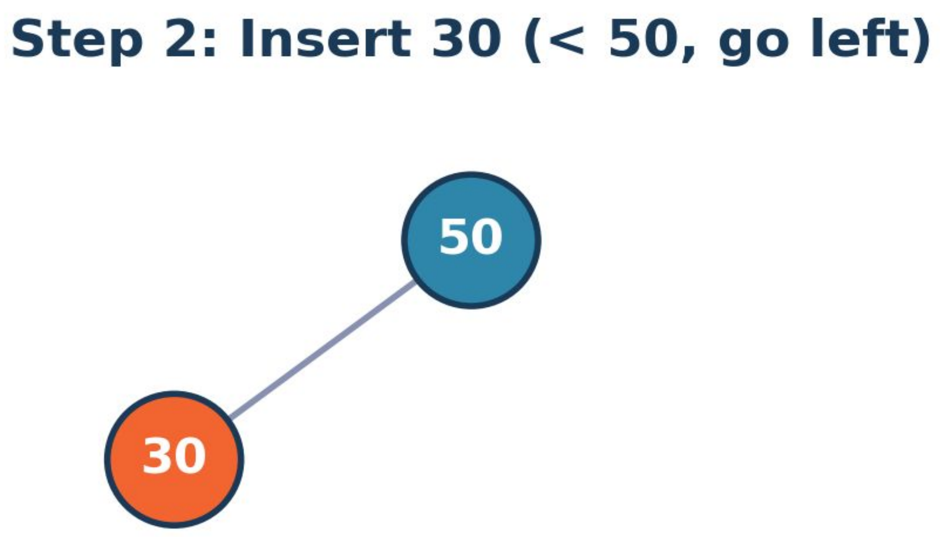
01

VisuAlgo

Animated BST & AVL Tree Visualizer

Renders animated binary search trees and AVL trees. Insert custom values and watch step-by-step operations — including rotations that keep AVL trees balanced.

- Custom value insertion
- Step-by-step animation controls
- BST & AVL modes side by side
- Free & widely used in CS courses



Illustrative: inserting a node into a BST

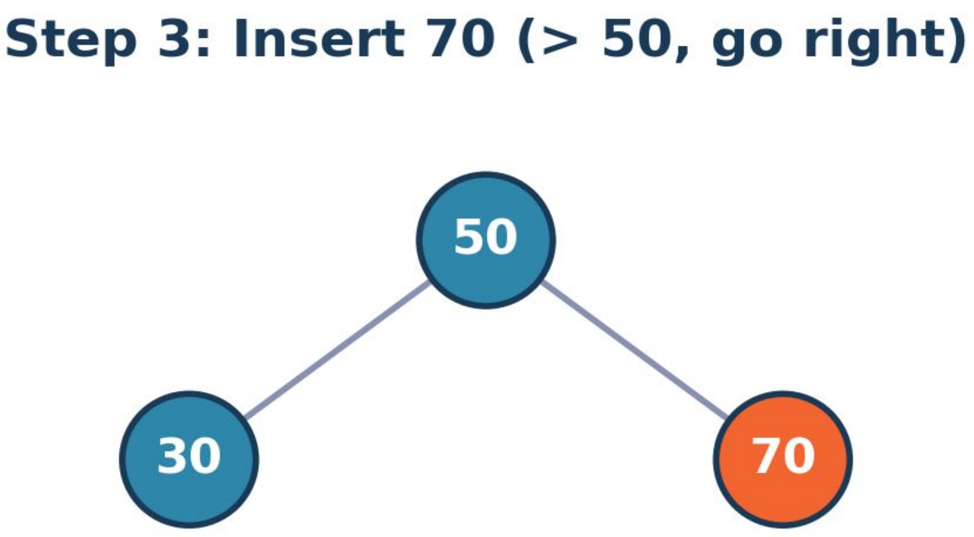
02

Binary Search Tree Visualizer

Hands-on Build, Search & Sort

Offers hands-on building, searching, and sorting visualizations for tree and graph structures, giving learners direct control over each operation.

- Interactive node building
- Live search highlighting
- Sorting via traversal
- Graph structure support



Illustrative: growing a search tree

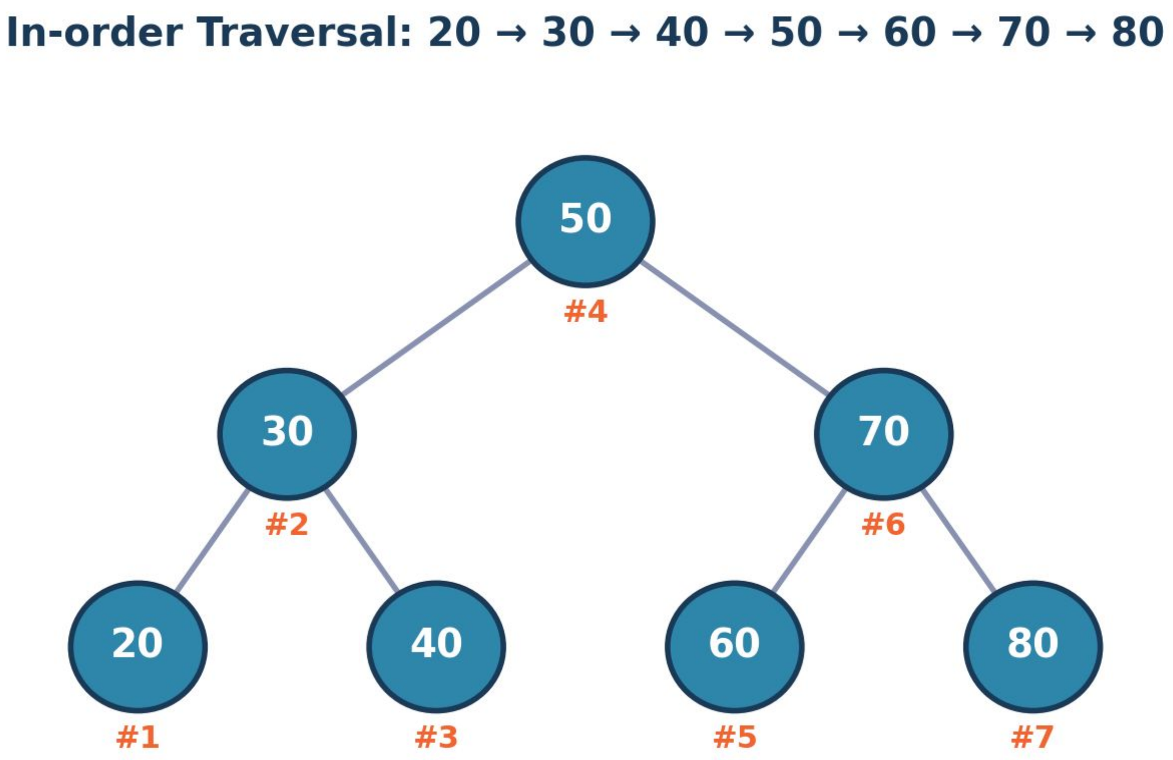
03

Binary Tree Visualizer & Converter

Traversal Order Tracing

Lets you input node lists and trace traversal orders visually against generated drawings, connecting raw data to the resulting tree shape.

- Node-list to tree conversion
- Pre/in/post-order tracing
- Visual traversal path
- Great for verifying homework



Illustrative: in-order traversal path

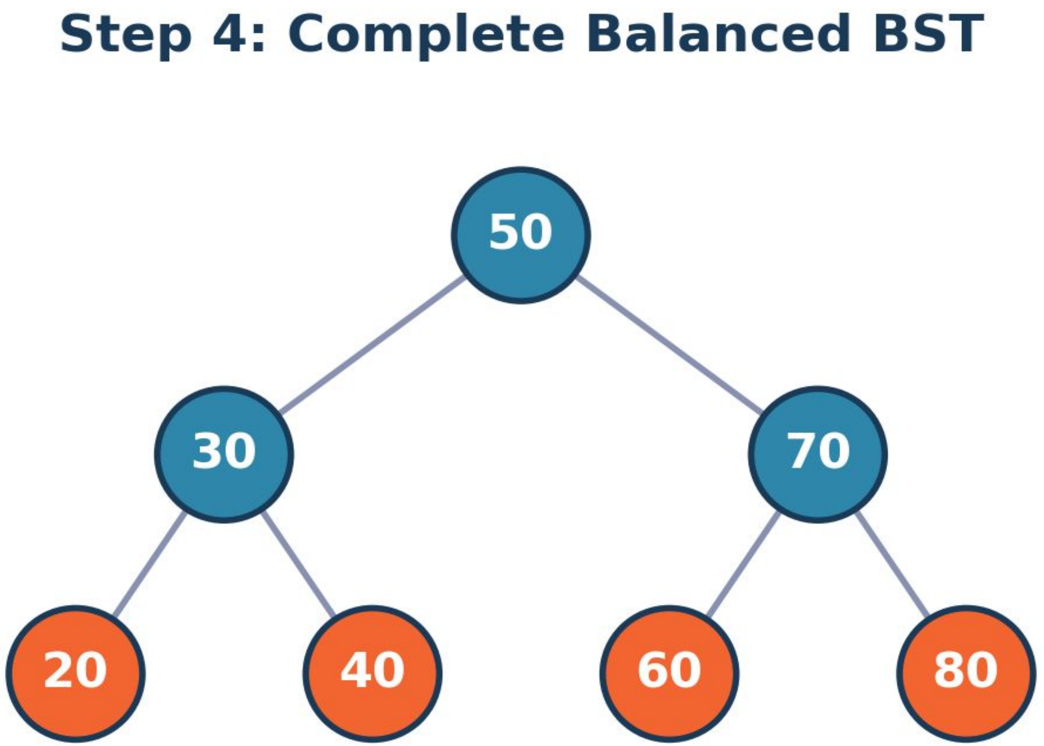
04

Virtual Labs IIITH

Tree Traversal Practice Module

A dedicated interactive practice module for clicking through nodes in the correct traversal order, with instant right/wrong feedback.

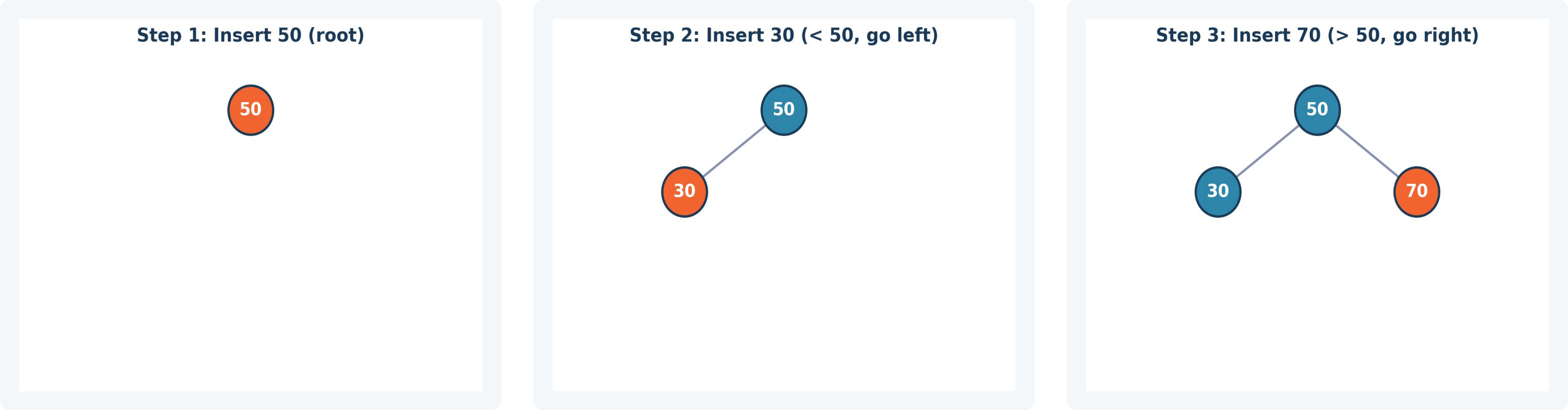
- Click-to-traverse practice
- Instant correctness feedback
- Structured lab exercises
- Backed by an academic institution



Illustrative: completed practice tree

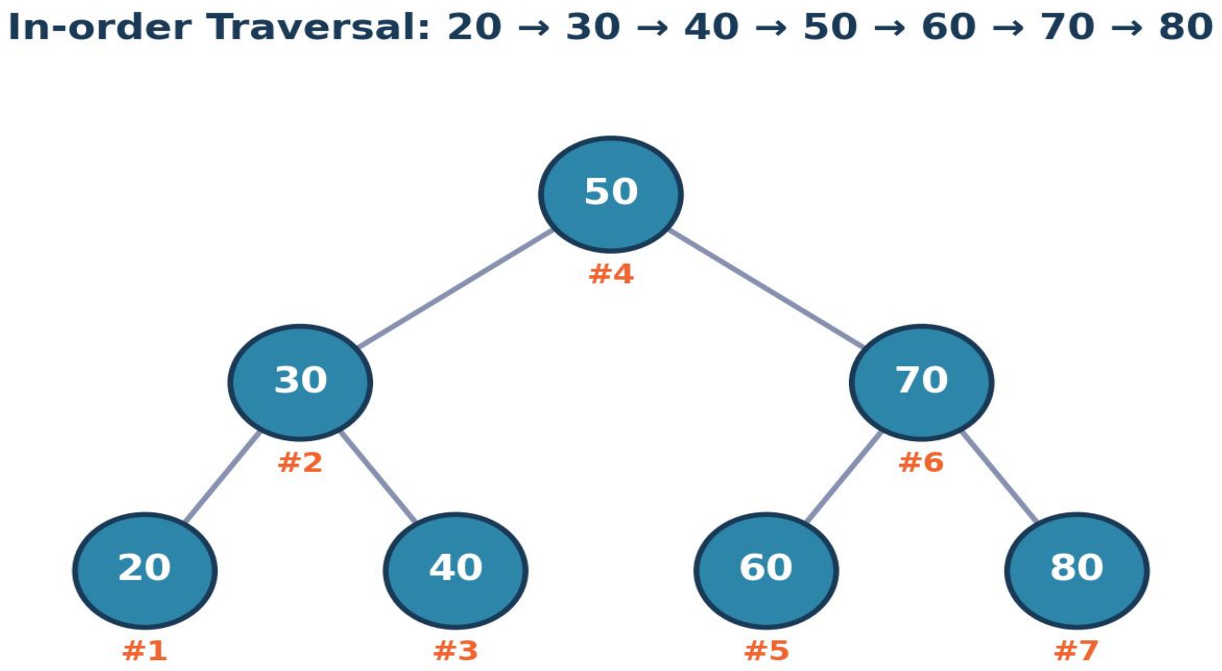
Frame-by-Frame BST Insertion

Inserting 50 ⟶ 30 ⟶ 70, one frame per step



Traversal Simulation Output

The same script tags each node with its visit order for any traversal type



Tool Comparison at a Glance

Tool	Best For	Interactivity	Access
VisuAlgo	BST & AVL animations	Insert / rotate step-by-step	Free, web-based
BST Visualizer	Build, search, sort	Full hands-on control	Free, web-based
Binary Tree Visualizer & Converter	Traversal tracing	Input node list, trace visually	Free, web-based
Virtual Labs IIITH	Traversal practice	Click-through with feedback	Academic lab portal
Our Frame-by-Frame Builder	Custom walkthroughs & handouts	Script-driven, fully customizable	Offline, Python-based

Binary Search Tree

Operations

- Insert
- Delete
- Find
- Traversal
- Min / Max
- Successor
- Predecessor
- Clear Tree

Animation Speed

Slow Medium Fast

Insert

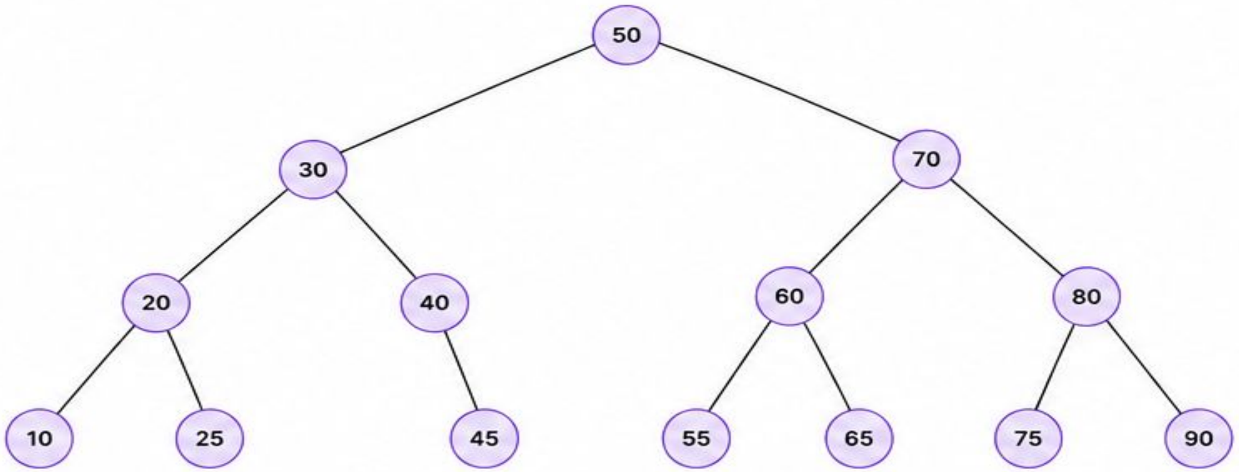
Enter value to insert

Enter integer value

Insert

Examples

Select example



Traversals

Inorder (LNR)
10, 20, 25, 30, 40, 45, 50, 55, 60, 65, 70, 75, 80, 90

Preorder (NLR)
50, 30, 20, 10, 25, 40, 45, 70, 60, 55, 65, 80, 75, 90

Postorder (LRN)
10, 25, 20, 45, 40, 30, 55, 65, 60, 75, 90, 80, 70, 50

Level Order
50, 30, 70, 20, 40, 60, 80, 10, 25, 45, 55, 65, 75, 90

Pause

Step: 12 / 12

Output / Steps

Inserting 65:

1. Start at root node 50
2. 65 > 50, move to right child 70
3. 65 > 70? No, move to left child 60
4. 65 > 60, move to right child 65
5. Inserted 65 as right child of 60

Insertion complete.

History

Insert 50
Insert 30
Insert 70
Insert 20
Insert 40
Insert 60
Insert 80
Insert 10
Insert 25
Insert 45
Insert 55
Insert 65